Antonio Cruciani

Aalto University (Espoo, Finland)

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https://antonio-cruciani.github.io/

Fields of Research Expertise

Distributed Computing, Graph Mining, Randomized Algorithms, Temporal Graphs, Data Mining, Approximation Algorithms, Statistical Learning

Current Position

Postdoctoral Researcher, Aalto University, Computer Science. Research Group of Prof. Jukka Suomela. *Espoo, FI* *Mar. 2025 – Present*

Education

Ph.D. in Computer Science, Gran Sasso Science Institute (GSSI). *L'Aquila, IT* *Mar. 2025*
Thesis: *Models and Algorithms for Temporal Betweenness Centrality and Dynamic Distributed Data Structures.* **Grade:** Excellent.
Supervisors: Francesco Pasquale (University of Rome "Tor Vergata"), Pierluigi Crescenzi (GSSI)
Reviewers: Christian Scheideler (UPB), Fabio Vandin (UNIPD).

M.S. in Computer Science, University of Rome "Tor Vergata", *Rome, IT* *2020*
Thesis: *Dynamic Random Graphs and unstructured P2P networks, analysis of two models inspired by the Bitcoin network.* Summa cum Laude.
Supervisor: Francesco Pasquale (University of Rome "Tor Vergata")

B.S. in Computer Science, University of Rome "Tor Vergata". *Rome, IT* *2017*
Thesis: *Efficient learning methods for playlist prediction.*
Supervisor: Giorgio Gambosi (University of Rome "Tor Vergata")

Visiting Research Periods

Visiting Researcher Fellow, Department of Computer Science and Engineering, University of Padua. *Padova, IT* *February 2026*
Host Professor: Leonardo Pellegrina

Visiting Researcher Fellow, KTH Royal Institute of Technology. *Stockholm, SE* *June 2025*
Host Professor: Aristides Gionis

Visiting Researcher Fellow, Department of Computer Science University of Hamburg. *Hamburg, GE* *January 2025*
Host Researcher: Thorsten Götte

Visiting Researcher Fellow, Department of Computer Science and Engineering, IIT Madras. *Chennai, IN* *1st-Aug. – 31st-Oct. 2024*
Host Professor: John Augustine

Visiting Researcher Fellow, Department of Computer Science and Engineering, University of Padua. *Padova, IT* *July 2024*
Host Professor: Leonardo Pellegrina

Visiting Researcher Fellow, Department of Computer Science and Engineering, IIT Madras. *Chennai, IN* *1st-Aug. 2023 – 27th-Feb. 2024*
Host Professor: John Augustine

Visiting Researcher Fellow, Big Data Analytics Lab, Fondazione Ugo Bordoni *Rome, IT*
Host Researcher: Giambattista Amati *Feb. – Nov. 2020*

Teaching Experience

Invited Teacher, *CS-E4565 - Combinatorics of Computation D.* (MSc), Aalto University (lecture delivery and guided problem solving, office hours, grading). **Topics:** probabilistic method, load balancing and concentration inequalities. *Finland, FI* *April - May. 2026*

Invited Teacher, *CS-E4565 - Combinatorics of Computation D.* (MSc), Aalto University (lecture delivery, guided problem solving, office hours, grading). **Topics:** introduction on probability, the probabilistic method, and derandomization across four sessions. *Finland, FI* *April - May. 2025*

Teaching Assistant, Computability and Computational Complexity Theory (BSc), University of Rome "Tor Vergata". (lecture delivery and guided problem solving) **Topics:** computability and computational complexity. *Rome, IT* *Oct. 2018 – Jun. 2019*

Teaching Assistant, Computer Programming in C (BSc), University of Rome "Tor Vergata". Led exercise sessions focused on problem solving and programming practice, supported students during in-class exercises, and graded weekly homework assignments. *Rome, IT* *Oct. 2017 – Jun. 2018*

Professional Experience

Software Developer, WeDot Rome. *Rome, IT* *Oct. 2015 – Jun. 2016*

Software Developer, New System. *Falerone, IT* *Jun. – Sep. 2010*

Languages

- English (fluent)
- Italian (mother tongue)

Programming Skills

- Basic: OWL, SPARQL, FORTRAN, COBOL, LISP
- Intermediate: GO, MATLAB, JAVASCRIPT, R, ASP.NET, PHP
- Advanced: PYTHON, JULIA, JAVA, C, C++,C#, SQL
- Frameworks: Apache Spark

Publications

In case of theoretical computer science conferences, authors are sorted alphabetically, otherwise by contribution.

Conference Proceedings

[1] **A. Cruciani**. "Maintaining a Bounded Degree Expander in Dynamic Peer-to-Peer Networks". In: *Structural Information and Communication Complexity - 33th International Colloquium, SIROCCO 2026, Durham, United Kingdom (TO APPEAR)*. Lecture Notes in Computer Science. 2026. Acceptance rate = 43%.

[2] **A. Cruciani** and L. Pellegrina. "Fast Percolation Centrality Approximation with Importance Sampling". In: *IEEE International Conference on Data Mining, ICDM 2025, Washington DC, USA, November 12-15, 2025*. IEEE, 2025, pp. 1125–1134. DOI: 10.1109/ICDM65498.2025.00121. URL: https://doi.org/10.1109/ICDM65498.2025.00121. Acceptance rate = 8.9%.

[3] J. Augustine, **A. Cruciani**, and I. A. Gillani. "Brief Announcement: Highly Dynamic and Fully Distributed Data Structures". In: *39th International Symposium on Distributed Computing (DISC 2025)*. Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 47:1–47:7. ISBN: 978-3-95977-402-4. DOI: 10.4230/LIPIcs.DISC.2025.47. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.47. Acceptance rate = 30.5%.

[4] A. Balliu, C. Coupette, **A. Cruciani**, F. d'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, and J. Suomela. "New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs". In: *39th International Symposium on Distributed Computing (DISC 2025)*. Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 11:1–11:22. ISBN: 978-3-95977-402-4. DOI: 10.4230/LIPIcs.DISC.2025.11. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.11. Acceptance rate = 30.5%.

[5] **A. Cruciani**. "Brief Announcement: Maintaining a Bounded Degree Expander in Dynamic Peer-To-Peer Networks". In: *39th International Symposium on Distributed Computing (DISC 2025)*. Ed. by D. R. Kowalski. Vol. 356. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 53:1–53:7. ISBN: 978-3-95977-402-4. DOI: 10.4230/LIPIcs.DISC.2025.53. URL: https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.DISC.2025.53. Acceptance rate = 30.5%.

[6] **A. Cruciani**. "MANTRA: Temporal Betweenness Centrality Approximation Through Sampling". In: *Machine Learning and Knowledge Discovery in Databases. Research Track - European Conference, ECML PKDD 2024, Vilnius, Lithuania, September 9-13, 2024, Proceedings, Part I*. Ed. by A. Bifet, J. Davis, T. Krilavicius, M. Kull, E. Ntoutsi, and I. Zliobaite. Vol. 14941. Lecture Notes in Computer Science. Springer, 2024, pp. 125–143. DOI: 10.1007/978-3-031-70341-6_8. URL: https://doi.org/10.1007/978-3-031-70341-6%5C_8. Acceptance rate = 24%.

[7] G. Amati, **A. Cruciani**, D. Pasquini, P. Vocca, and S. Angelini. "propagate: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs". In: *Machine Learning and Knowledge Discovery in Databases: Research Track - European Conference, ECML PKDD 2023, Turin, Italy, September 18-22, 2023, Proceedings, Part III*. Ed. by D. Koutra, C. Plant, M. G. Rodriguez, E. Baralis, and F. Bonchi. Vol. 14171. Lecture Notes in Computer Science. Springer, 2023, pp. 671–688. DOI: 10.1007/978-3-031-43418-1_40. URL: https://doi.org/10.1007/978-3-031-43418-1%5C_40. Acceptance rate = 24%.

[8] R. Becker, P. Crescenzi, **A. Cruciani**, and B. Kodric. "Proxying Betweenness Centrality Rankings in Temporal Networks". In: *21st International Symposium on Experimental Algorithms, SEA 2023, July 24-26, 2023, Barcelona, Spain*. Ed. by L. Georgiadis. Vol. 265. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023, 6:1–6:22. DOI: 10.4230/LIPICS.SEA.2023.6. URL: https://doi.org/10.4230/LIPIcs.SEA.2023.6. Acceptance rate = 50%.

[9] **A. Cruciani** and F. Pasquale. "Dynamic graph models inspired by the Bitcoin network-formation process". In: *24th International Conference on Distributed Computing and Networking, ICDCN 2023, Kharagpur, India, January 4-7, 2023*. ACM, 2023, pp. 125–134. DOI: 10.1145/3571306.3571398. URL: https://doi.org/10.1145/3571306.3571398. Acceptance rate = 34%.

[10] **A. Cruciani** and F. Pasquale. "Brief Announcement: Dynamic Graph Models for the Bitcoin P2P Network: Simulation Analysis for Expansion and Flooding Time". In: *Stabilization, Safety, and Security of Distributed Systems - 24th International Symposium, SSS 2022, Clermont-Ferrand, France, November 15-17, 2022, Proceedings*. Ed. by S. Devismes, F. Petit, K. Altisen, G. A. D. Luna, and A. F. Anta. Vol. 13751. Lecture Notes in Computer Science. Springer, 2022, pp. 335–340. DOI: 10.1007/978-3-031-21017-4_23. URL: https://doi.org/10.1007/978-3-031-21017-4%5C_23.

[11] G. Amati, S. Angelini, **A. Cruciani**, G. Fusco, G. Gaudino, D. Pasquini, and P. Vocca. "Topic Modeling by Community Detection Algorithms". In: *OASIS@HT 2021: Proceedings of the 2021 Workshop on Open Challenges in Online Social Networks, Virtual Event, Ireland, 30 August 2021*. Ed. by B. Guidi, A. Michienzi, and L. Ricci. ACM, 2021, pp. 15–20. DOI: 10.1145/3472720.3483622. URL: https://doi.org/10.1145/3472720.3483622.

[12] **A. Cruciani**, D. Pasquini, G. Amati, and P. Vocca. "About Graph Index Compression Techniques". In: *Proceedings of the 10th Italian Information Retrieval Workshop, Padova, Italy, September 16-18, 2019*. Ed. by M. Agosti, E. D. Buccio, M. Melucci, S. Mizzaro, G. Pasi, and F. Silvestri. Vol. 2441. CEUR Workshop Proceedings. CEUR-WS.org, 2019, pp. 21–24. URL: https://ceur-ws.org/Vol-2441/paper23.pdf.

Preprints

[13] J. Augustine, **A. Cruciani**, and I. A. Gillani. *Highly Dynamic and Fully Distributed Data Structures*. 2025. arXiv: 2409.10235 [cs.DC]. URL: https://arxiv.org/abs/2409.10235.

[14] **A. Cruciani**, A. Das, A. Raevskaya, and J. Suomela. *It does not matter how you define locally checkable labelings*. 2026. arXiv: 2602.18188 [cs.DC]. URL: https://arxiv.org/abs/2602.18188.

[15] **A. Cruciani**, A. Das, M. Equi, H. Lievonen, D. Luong-Le, A. Modanese, and J. Suomela. *Is a LOCAL algorithm computable?* 2026. arXiv: 2602.21022 [cs.DC]. URL: https://arxiv.org/abs/2602.21022.

PhD Thesis

[16] **A. Cruciani**. *Highly Dynamic and Fully Distributed Data Structures*. 2025. URL: https://iris.gssi.it/handle/20.500.12571/34764.

Presentations at Conferences & Workshops

- **2025:**
  - Fast Percolation Centrality Approximation with Importance Sampling (ICDM, Washington DC, USA);
  - Maintaining a Bounded Degree Expander in Dynamic Peer-To-Peer Networks (DISC, Berlin, DE);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (DISC, Berlin, DE);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (Joint Estonian-Latvian Theory Days, Riga, LV);
  - Fast Percolation Centrality Approximation with Importance Sampling (Helsinki Algorithms & Theory Days, Helsinki, FI).
- **2024:**
  - Approximating Distance-based metrics through sampling (IIT Madras, Chennai, IN);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (Online, Aalto University, Espoo, FI);
  - Maintaining Distributed Data Structures in Dynamic Peer-to-Peer Networks (IIT Madras, Chennai, IN);
  - On the Temporal Betweenness Centrality (IIT Madras, Chennai, IN);
  - MANTRA: Temporal Betweenness Centrality Approximation through Sampling (ECML-PKDD, Vilnius, LT);
  - Computing Distance-based metrics on Very Large Graphs (UNIPD, Padova, IT).
- **2023:**
  - PROPAGATE: A Seed Propagation Framework to Compute Distance-Based Metrics on Very Large Graphs (ECML-PKDD, Turin, IT);
  - Proxying Betweenness Centrality Rankings in Temporal Networks (SEA, Barcelona, ES);
  - Dynamic graph models inspired by the Bitcoin network-formation process (ICDCN, Kharagpur, IN).
- **2022:**
  - Dynamic graph models for the Bitcoin P2P network: simulation analysis for expansion and flooding time (SSS, Clermont-Ferrand, FR).
- **2019:**
  - About Graph Index Compression Techniques (IIR, Padova, IT);
  - Iterative Compression technique for NP-Hard problems on Graphs (University of Rome Tor Vergata, Rome, IT).

Additional Training

Bertinoro International Spring School	*Bertinoro, IT* *Mar. 2022*
European Summer School on Learning in Games, Markets, and Online Decision Making	*Rome, IT* *Sep. 2021*
Max Planck Advanced Course on the Foundations of Computer Science (Convex Optimization) (online)	*Saarbrucken, GE* *Jul. - Aug. 2021*
Algorithmic Tools for Massive Network Analytics (online)	*Pisa, IT* *May - Jun. 2021*
Max Planck Advanced Course on the Foundations of Computer Science (Market Design and Computational Fair Division) (online)	*Saarbrucken, GE* *August 2020*
Algorithms and computational models for large-scale data analysis. University of Rome: "La Sapienza". By Silvio Lattanzi (Google Research).	*Rome, IT* *August 2019*

Academic Service

Conferences

PC-Member SSS 2026
Reviewer ICALP, PODC 2026
Reviewer STACS-PODC-WSDM-SDM-SEA-ISAAC-FSTTCS, ICANN 2025
Reviewer FUN 2020-2024
Reviewer AAMAS 2023

Journals

Reviewer: The Review of Socionetwork Strategies

Supervision

Bachelor Students: Giacomo Rivetti (2025, co-supervised with Francesco Pasquale)

Software Packages

- PROPAGATE, an efficient algorithm for approximating various distance-based metrics (i.e., average distance, effective diameter, diameter and connectivity rate).
- DREG, a dynamic expander graph generator.
- TSBProxy, a suite of efficient proxies for the temporal betweenness centrality rankings.
- MANTRA, an efficient framework for approximating the temporal betweenness centrality using sampling.
- FEPIC, an efficient approximation algorithm for the (doubly normalized) percolation centrality.
- PercIS, a fast approximation algorithm for the percolation centrality that uses importance sampling.

Contact References

- John Augustine, IIT Madras: augustine@cse.iitm.ac.in
- Jukka Suomela, Aalto University: jukka.suomela@aalto.fi
- Francesco Pasquale, University of Rome Tor Vergata: pasquale@mat.uniroma2.it
- Leonardo Pellegrina, University of Padua: leonardo.pellegrina@unipd.it
- Giambattista Amati, Fondazione Ugo Bordoni: gba@fub.it
- Thorsten Götte, Hamburg University: thorsten.goette@uni-hamburg.de